

Name: _____



Exit quiz

Problem solving with sequences

1 Which would describe the sequence 1,3,6,10,15, ...? Tick **1** correct answer

☐ Arithmetic

☐ Linear

☐ Non-linear

2 The 2nd term of an arithmetic sequence is 11 and the 6th term is 47. What is the 1st term of the sequence? Tick **1** correct answer

☐ 0

☐ 1

☐ 2

☐ 3

☐ 4

3 Buses leave Wallsend for Byker at regular intervals. The second bus of the day leaves at 6:44 am. You just missed the seventh bus of the day which left at 8:04 am. What time is the next bus? Tick **1** correct answer

☐ 8:14 am

☐ 8:20 am

☐ 8:24 am

☐ 8:30 am

☐ 8:34 am



- 4 The 24th term of an arithmetic sequence is 281. The 37th term is 437. Which calculation will be the starting point to finding the n^{th} term of the sequence? Tick **1** correct answer

☐ $\frac{437 - 281}{37 - 24}$

☐ $\frac{437 - 281}{24 - 37}$

☐ $\frac{281 - 437}{37 - 24}$

☐ $\frac{37 - 24}{437 - 281}$

☐ $\frac{24 - 37}{437 - 281}$

- 5 What is the n^{th} term of this sequence? Tick **1** correct answer

$$\frac{2}{3}, \frac{9}{8}, \frac{16}{13}, \frac{23}{18}, \dots$$

☐ $\frac{n + 7}{n + 5}$

☐ $\frac{7n + 2}{5n + 3}$

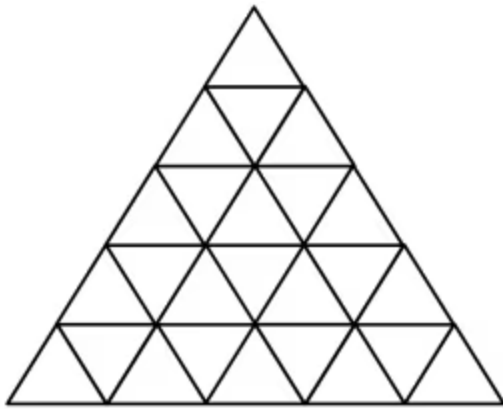
☐ $\frac{7n + 5}{5n + 2}$

☐ $\frac{7n - 5}{5n - 2}$

☐ It's not an arithmetic (linear) sequence we can't write an n^{th} term.



- 6 How many triangles are there in this pattern? Don't forget to count all triangles of all sizes.
Tick **1** correct answer

☐ 50☐ 48☐ 45☐ 38☐ 35